BTS–V(S)–04.18–0766 Reg. No. [ ] **A**

B.Tech. Degree V Semester Supplementary Examination April 2018

ME 15-1504 THERMAL ENGINEERING

(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

(Appropriate charts can be used. Assume missing data if any suitably)

PART A

(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) Explain the ultimate and proximate analysis of coal.

(b) Explain adiabatic flame temperature and theoretical air for combustion.

(c) Explain second law of thermodynamics for reactive systems.

(d) Explain why we need "Air-standard" cycle analysis for IC engines. State assumptions made.

(e) Draw actual valve timing diagram for 4 stroke cycle petrol engine and explain the valve overlap.

(f) Using the T-S diagram, prove that, for the same quantity of heat added, increase of compression ratio increases the thermal efficiency of Otto cycles.

(g) What is meant by supercharging in IC engines? Explain.

(h) Explain clearly the function of condenser in the ignition system.

(i) Explain pre-ignition and auto-ignition in IC engines.

(j) What is delay period? What are the factors that affect it?

PART B

$(4 \times 10 = 40)$

II. (a) Explain with neat sketch the working of Bomb calorimeter. (4)

(b) The reaction equation of a fuel is represented by (6)

$C_nH_m + aO_2 + bN_2 = 8CO_2 + 0.9CO + 8.8O_2 + dH_2O + 82.3\ N_2$.

Determine (i) The actual air – fuel ratio and the chemical formula of the fuel (ii) The stoichiometric air – fuel ratio and percentage theoretical air used. Assume $N_2 : O_2$ *is* 3.76 : 1.

OR

III. (a) Explain with neat sketch the analysis of combustion products using Orsat apparatus. (4)

(b) A gasoline engine delivers 150 kW. The fuel used is liquid octane and air enters at 45°C. The products of combustion leave the engine at 750K and the heat transfer from the engine is 205kW. Determine the fuel consumption per hour if complete combustion is achieved with 50% excess air. (6)

(P.T.O.)

IV. (a) What would be the percentage change in the ideal efficiency of a diesel engine having a compression ratio 14 when the fuel cut-off is delayed from 5% to 8%? (5)

(b) An ideal air engine having a compression ratio 'r' sucks in atmospheric air at 105kPa and 300K. The maximum permissible pressure and temperature inside the engine is 8000kPa and 3000 K. Estimate the mass of air trapped in the clearance space and the compression ratio. Take $\gamma = 1.4$ *and* $R = 287 J/kg\,K$. (5)

OR

V. (a) In an air standard dual combustion cycle with a compression ratio of 10, the minimum pressure and temperature are 1 bar and 288 K respectively. The maximum temperature and pressure are 2400K and 50 bar, respectively. Determine the work done per kg of air, the energy added as heat per kg of air and the thermal efficiency of the cycle. (5)

(b) A four cylinder, four stroke petrol engine, 80 mm bore, 120 mm stroke develops 27.4kW brake power which is running at 1500 rpm and using a rich mixture. If the volume of air in to the cylinder, when measured at 16.2°C and 760 mm of mercury is 70% swept volume, the theoretical A:F is 14.7:1, the heating value of fuel used is 43000kJ/kg and the mechanical efficiency of the engine is 89%, find the indicated thermal efficiency. Take R=0.287kJ/kgK. (5)

VI. (a) Explain the components of transmission system in IC engines. (5)

(b) Explain with the help of a simple sketch the working of pressure feed lubrication systems. List out its merits and demerits. (5)

OR

VII. (a) What is the purpose of Governing of IC engines? Describe briefly the working of quantity governing of IC engines. List out its merits and demerits. (5)

(b) Classify the injection systems in diesel engines. With a neat sketch explain the working of a diesel fuel injector. (5)

VIII. (a) How are SI engine fuels rated? What do you understand by Octane number – 110? (5)

(b) What are the different types of combustion chambers used in CI engines? Explain any two with a neat sketch. What are the reasons for operating a CI engine at considerably higher A:F ratio compared to chemically correct mixture. (5)

OR

IX. (a) Explain the following related to internal combustion engines (i) Abnormal combustion. (ii) Un-controlled combustion (iii) Diesel knock (iv) Cetane number (v) Run – on and Run – away. (5)

(b) List out the requirements of a good combustion chamber for S.I engines. Also differentiate the advantages of L-type combustion chambers with OHV chambers used for SI engines. (5)

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